4 Maisie knows that she needs 3 kg of grass seed to make a rectangular lawn 5 m by 9 m.

Grass seed is sold in 2 kg boxes.

Maisie wants to make a rectangular lawn 10 m by 14 m.
She has 5 boxes of grass seed.

(a) Has Maisie got enough grass seed to make a lawn 10 m by 14 m?
You must show all your working.

Original area = 5 × 9 = 45 m²; 3 kg covers 45 m².

New area = 10 × 14 = 140 m².

Seed needed = 3 × 140/45 = 28/3 ≈ 9.33 kg.

Five 2 kg boxes contain 10 kg, so yes: enough.

4 marks total
Compare areas
Scale 3 kg to 140 m²
Need 28/3 kg
10 kg: yes

(4)

Maisie opens the 5 boxes of grass seed.

She finds that 4 of the boxes contain 2 kg of grass seed.
The other box contains 1 kg of grass seed.

(b) Does this affect whether Maisie has enough grass seed to make her lawn?
Give a reason for your answer.

Now total seed = 4 × 2 + 1 = 9 kg.

9 kg < 28/3 kg needed, so she no longer has enough.

1 mark
9 kg < 28/3 kg
This changes the answer

(1)

(Total for Question 4 is 5 marks)

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